this is about a million billion times the mass of the proton, or about four orders of magnitude less than the Planck mass. Georgi and Glashow boldly took theoretical physics into an energy realm many orders of magnitude beyond that which anyone had previously dared explore.

Subsequent work at Harvard by Georgi, Helen Quinn, and Weinberg in 1974 made the potential unity of the nongravitational forces within the grand unified framework even more manifest. As their contribution continues to play an important role in unifying the forces and in assessing the relevance of supersymmetry to the natural world, let’s spend a moment explaining it.

We are all aware that the electrical attraction between two oppositely charged particles or the gravitational attraction between two massive bodies gets stronger as the distance between the objects decreases. These are simple and well-known features of classical physics. There is a surprise, though, when we study the effect that quantum physics has on force strengths. Why should quantum mechanics have any effect at all? The answer, once again, lies in quantum fluctuations. When we examine the electric force field of an electron, for example, we are actually examining it through the “mist” of momentary particle-antiparticle eruptions and annihilations that are occurring all through the region of space surrounding it. Physicists some time ago realized that this seething mist of microscopic fluctuations obscures the full strength of the electron’s force field, somewhat as a thin fog partially obscures the beacon of a lighthouse. But notice that as we get closer to the electron, we will have penetrated more of the cloaking particle-antiparticle mist and hence will be less subject to its diminishing influence. This implies that the strength of an electron’s electric field will increase as we get closer to it.

Physicists distinguish this quantum-mechanical increase in strength as we get closer to the electron from that known in classical physics by saying that the intrinsic strength of the electromagnetic force increases on shorter distance scales. This reflects that the strength increases not merely because we are closer to the electron but also because more of the electron’s intrinsic electric field becomes visible. In fact, although we have focused on the electron, this discussion applies equally well to all electrically charged particles and is summarized by saying that